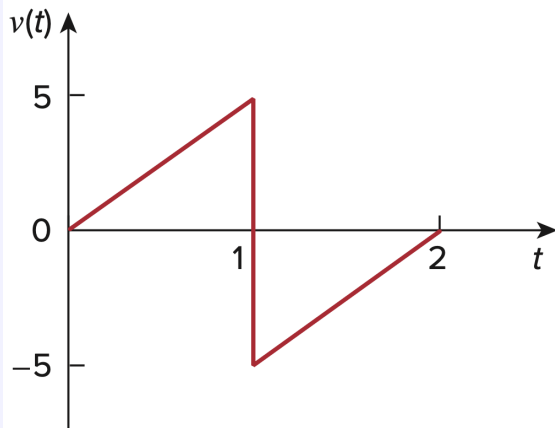


Exercise Sheet #5

Fields 2

Exercise 1: Time-Dependent Inductor Voltage and Current

If the voltage waveform below is applied to a 10-mH inductor, find the inductor current $i(t)$ for $0 < t < 2$ seconds. Assume $i(0) = 0$.



Solution

For an inductor:

$$v(t) = L \frac{di(t)}{dt}$$

$0 < t < 1$ s (Ramp up $0 \rightarrow 5$ V):

$$v(t) = 5t \text{ V} \quad \Rightarrow \quad \frac{di}{dt} = \frac{5t \text{ V}}{0.01 \text{ H}} = 500t \text{ A/s}$$

Integrating:

$$i(t) = \int 500t \text{ A/s } dt = (250t^2 + C) \text{ A}$$

Since $i(0) = 0$, $C = 0$:

$$i(t) = 250t^2 \text{ A}, \quad 0 < t < 1$$

$t = 1$ s (instantaneous drop $5 \rightarrow -5$ V):

An ideal inductor cannot change current instantaneously. Therefore, the inductor current is continuous at $t = 1$:

$$i(1^-) = i(1^+) = 250 \text{ A}$$

The instantaneous voltage jump produces a finite slope change in di/dt , $i(t)$ itself is continuous.

$1 < t < 2$ s (ramp from $-5 \rightarrow 0$ V):

Voltage is now

$$v(t) = -5 \text{ V} + 5(t - 1) \text{ V} = (5t - 10) \text{ V}, \quad 1 < t < 2.$$

Then

$$\frac{di}{dt} = \frac{(5t - 10) \text{ V}}{0.01 \text{ H}} = (500t - 1000) \text{ A/s.}$$

Integrate:

$$i(t) = \int (500t - 1000) \text{ A/s} dt = (250t^2 - 1000t + C) \text{ A}$$

Use continuity at $t = 1$: $i(1) = 250 \text{ A}$

$$250 \text{ A} = (250(1)^2 - 1000(1) + C) \text{ A} \implies C = 1000$$

So:

$$i(t) = (250t^2 - 1000t + 1000) \text{ A}, \quad 1 < t < 2$$

Exercise 2: Coil

A cylindrical coil with a length of 2.7 m, a radius of 0.85 cm, and 600 turns carries a current of 2.5 A. Calculate B inside the coil, far from its ends.

Solution

The magnetic field inside a long solenoid (far from its ends) is given by

$$B = \mu_0 n I,$$

where n is the number of turns per unit length:

$$n = \frac{N}{l} = \frac{600}{2.7 \text{ m}} \approx 222.22 \text{ turns/m}$$

With the permeability of free space

$$\mu_0 = 4\pi \cdot 10^{-7} \text{ H/m},$$

the magnetic field is:

$$B = (4\pi \cdot 10^{-7}) \cdot 222.22 \cdot 2.5 \text{ T} = 6.98 \cdot 10^{-4} \text{ T} \approx 0.7 \text{ mT}$$

Exercise 3: Energy of Inductors

The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

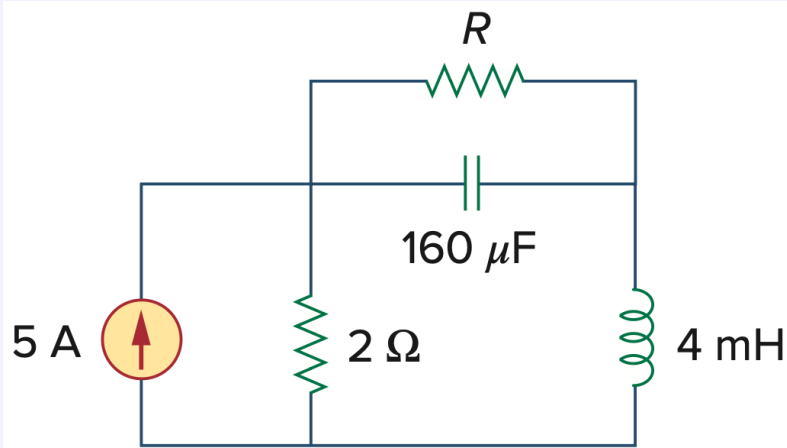
Solution

The energy stored in the inductor is given by

$$W = \frac{1}{2} L I^2 = \frac{1}{2} (0.08) (0.06)^2 \text{ J} = 144 \text{ } \mu\text{J}.$$

Exercise 4: Inductor and Capacitor under DC

For the circuit below, calculate the value of R that will make the energy stored in the capacitor the same as that stored in the inductor under DC conditions.



Solution

Under steady-state DC, the capacitor is an open circuit and the inductor is a short circuit. Call the top node voltage V . The $2\text{-}\Omega$ resistor and R both go to ground, so

$$\frac{V}{R} + \frac{V}{2\Omega} = 5\text{ A} \Rightarrow V = \frac{10R}{R+2}.$$

The inductor DC current is the current through R :

$$I_L = \frac{V}{R} = \frac{10}{R+2}$$

Equate stored energies:

$$\frac{1}{2}CV^2 = \frac{1}{2}LI_L^2$$

Substituting V and I_L gives

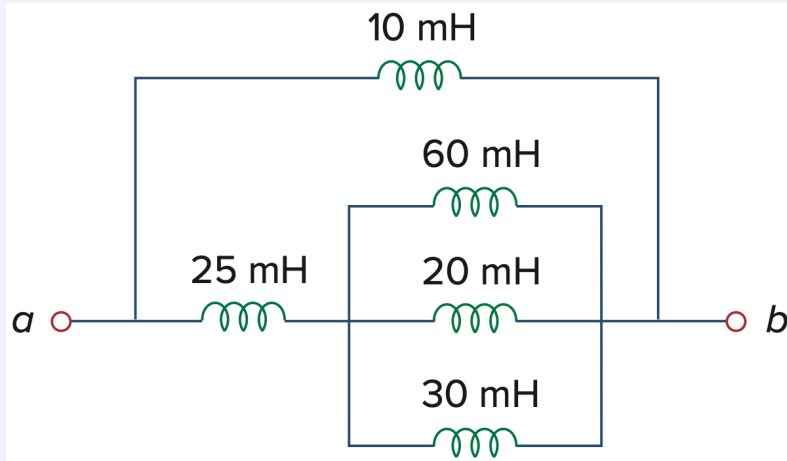
$$C\left(\frac{10R}{R+2}\right)^2 = L\left(\frac{10}{R+2}\right)^2 \Rightarrow CR^2 = L,$$

and therefore

$$R = \sqrt{\frac{L}{C}} = \sqrt{\frac{4 \times 10^{-3}}{160 \times 10^{-6}}} \Omega = 5 \Omega.$$

Exercise 5: Series and Parallel Inductors

Determine L_{eq} at terminals a-b of the circuit below.



Solution

Step 1: Parallel combination of the three inductances with 60 mH, 20 mH, 30 mH:

$$\frac{1}{L_p} = \frac{1}{60 \text{ mH}} + \frac{1}{20 \text{ mH}} + \frac{1}{30 \text{ mH}} = \frac{1}{10} \frac{1}{\text{mH}}$$

$$\Rightarrow L_p = 10 \text{ mH}$$

Step 2: Series combination with the inductance of 25 mH:

$$L_s = L_p + 25 \text{ mH} = 10 \text{ mH} + 25 \text{ mH} = 35 \text{ mH}$$

Step 3: Parallel combination with the inductance of 10 mH:

$$\frac{1}{L_{eq}} = \frac{1}{L_s} + \frac{1}{10 \text{ mH}} = \frac{1}{35 \text{ mH}} + \frac{1}{10 \text{ mH}} = \frac{9}{70} \frac{1}{\text{mH}}$$

$$\Rightarrow L_{eq} \approx 7.78 \text{ mH}$$

Exercise 6: Ideal Transformer

A 480-V to 2400-V step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

- the turns ratio
- the primary current
- the secondary current

Solution

a) Turns ratio (primary to secondary):

$$\alpha = \frac{V_P}{V_S} = \frac{480 \text{ V}}{2400 \text{ V}} = 0.2$$

b) Primary current:

$$I_P = \frac{P}{V_P} = \frac{50000 \text{ W}}{480 \text{ V}} \approx 104.17 \text{ A}$$

c) Secondary current:

$$I_S = \frac{P}{V_S} = \frac{50000 \text{ W}}{2400 \text{ V}} \approx 20.83 \text{ A}$$

Check:

$$\frac{I_S}{I_P} \approx 0.2$$

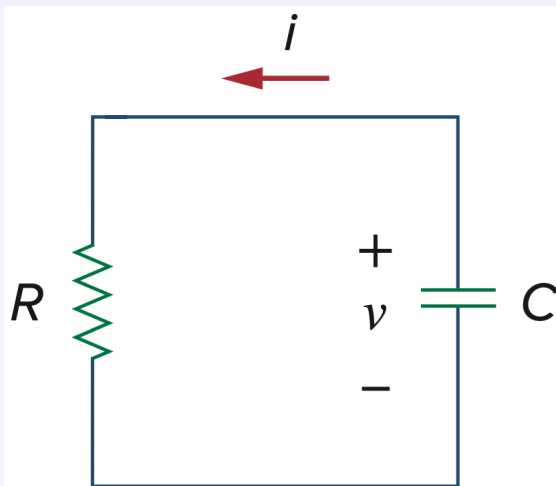
Remark: For AC, one would simply consider RMS values.

Exercise 7: RC-Discharge

In the circuit shown below, for $t > 0$

$$v(t) = 56e^{-200t} \text{ V}, \quad i(t) = 8e^{-200t} \text{ mA}.$$

- Find the values of R and C .
- Calculate the time constant τ .
- Determine the time required for the voltage to decay to half its initial value at $t = 0$.



Solution

- a) For a single series R – C discharge, the capacitor voltage has the form

$$v(t) = V_0 e^{-t/(RC)}.$$

Comparing $v(t) = 56e^{-200t}$ with $V_0 e^{-t/(RC)}$ gives

$$\frac{1}{RC} = 200 \implies RC = 5 \text{ ms}.$$

The capacitor current is related to the capacitor voltage by

$$i(t) = C \frac{dv(t)}{dt}.$$

Differentiate $v(t)$:

$$\frac{dv}{dt} = -200 \cdot 56e^{-200t} \text{ V/s} = -11200e^{-200t} \text{ V/s}.$$

Hence

$$C = \frac{i(t)}{dv/dt} = \frac{0.008e^{-200t}}{-11200e^{-200t}}.$$

Taking the positive magnitude,

$$C = \frac{0.008}{11200} \text{ F} \approx 714.3 \text{ nF}.$$

$$R = \frac{RC}{C} = \frac{0.005}{714.3 \times 10^{-9}} \Omega = 7 \text{ k}\Omega$$

(Alternatively, $R = v(t)/i(t) = 56/0.008 \Omega = 7 \text{ k}\Omega$.)

b) The time constant is

$$\tau = RC = 5 \text{ ms}.$$

c) We require $v(t_{1/2}) = \frac{1}{2}V_0$. For $v(t) = V_0e^{-t/\tau}$,

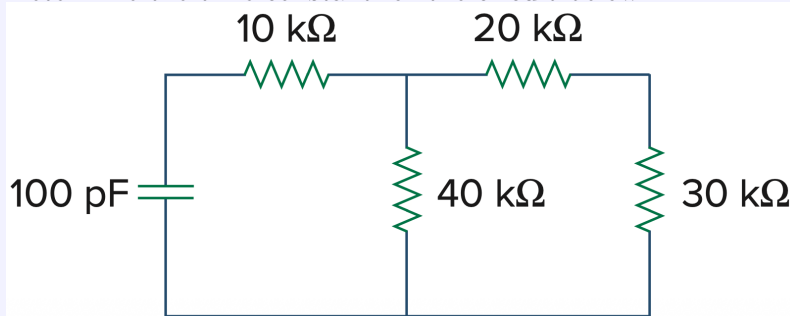
$$\frac{1}{2} = e^{-t_{1/2}/\tau} \implies t_{1/2} = \tau \ln 2.$$

Thus

$$t_{1/2} = (0.005 \cdot \ln 2) \text{ s} \approx 3.466 \text{ ms}.$$

Exercise 8: Time Constant

Determine the time constant for the circuit below.



Solution

Find the Thevenin resistance:

$$R_{\text{th}} = 10 \text{ k}\Omega + (40 \text{ k}\Omega \parallel (20 \text{ k}\Omega + 30 \text{ k}\Omega)) = 10 \text{ k}\Omega + \frac{40 \cdot 50}{40 + 50} \text{ k}\Omega = \frac{290}{9} \text{ k}\Omega$$

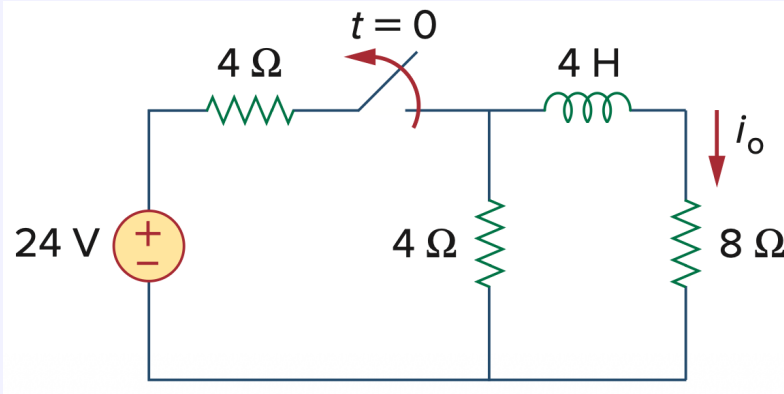
Compute the time constant:

$$\tau = R_{\text{th}}C = \left(\frac{290}{9} \times 10^3 \Omega\right) (100 \times 10^{-12} \text{ F}) = \frac{29000}{9} \times 10^{-9} \text{ s}$$

$$\tau \approx 3.22 \mu\text{s}$$

Exercise 9: Inductor Current Decay

For the circuit below, find i_o for $t > 0$.

**Solution**

The switch is closed for $t < 0$. Therefore, the inductor is in DC steady state and behaves as a short circuit.

The node after the left 4Ω resistor sees the parallel of 4Ω and 8Ω :

$$R_{\text{eq}} = \frac{4 \cdot 8}{4 + 8} = \frac{8}{3} \Omega.$$

The node voltage is

$$v(0^-) = 24 \frac{R_{\text{eq}}}{4 + R_{\text{eq}}} = 24 \frac{8/3}{4 + 8/3} = 9.6 \text{ V}.$$

Thus the inductor current at $t = 0^-$ is

$$i_L(0^-) = \frac{v(0^-)}{8} = \frac{9.6}{8} = 1.2 \text{ A}.$$

Therefore,

$$i_L(0^+) = 1.2 \text{ A}.$$

After the switch opens at $t = 0$, the source is disconnected. The top node is connected only to a 4Ω resistor to ground and the inductor with $L = 4\text{ H}$ in series with an 8Ω resistor.

Applying KCL at the top node (currents downward positive):

$$\frac{v}{4} + i_L = 0$$

Hence,

$$v = -4i_L.$$

The right branch voltage is also

$$v = 8i_L + L \frac{di_L}{dt} = 8i_L + 4 \frac{di_L}{dt}.$$

Equate the two expressions for v :

$$-4i_L = 8i_L + 4 \frac{di_L}{dt}$$

Divide by 4:

$$\frac{di_L}{dt} + 3i_L = 0.$$

Thus,

$$i_L(t) = i_L(0^+) e^{-3t} = 1.2 e^{-3t}.$$

$$\boxed{i_o(t) = 1.2 e^{-3t} \text{ A}, \quad t > 0.}$$

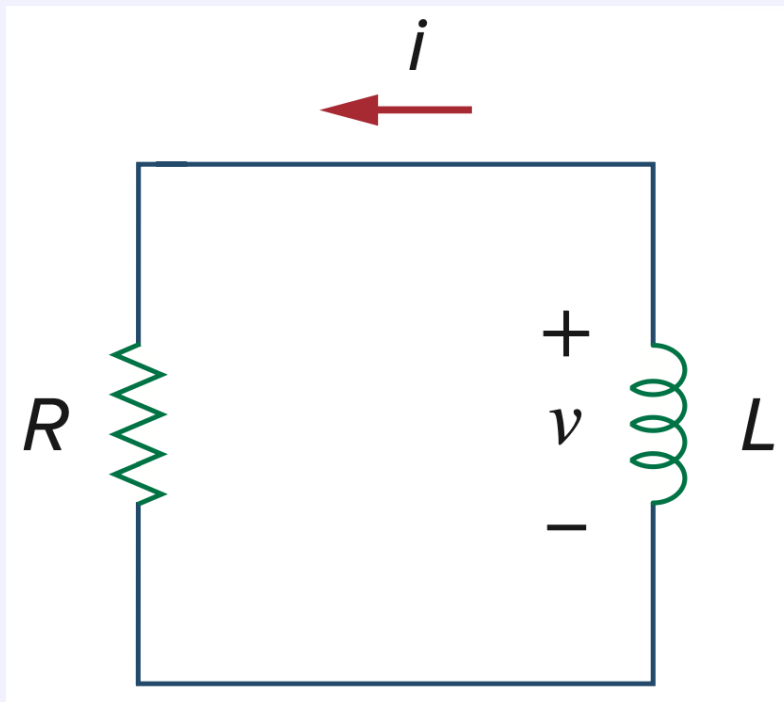
Exercise 10: Energy Dissipation during Transient

In the circuit below,

$$v(t) = 80e^{-10^3 t} \text{ V}, \quad t > 0,$$

$$i(t) = 5e^{-10^3 t} \text{ mA}, \quad t > 0.$$

- Find R , L , τ .
- Calculate the energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.



Solution

- The current has the form $i(t) = I_0 e^{-t/\tau}$. Comparing exponents,

$$\frac{1}{\tau} = 1000 \implies \tau = 1 \text{ ms}.$$

Since $v(t)$ is also the voltage across the resistor,

$$R = \frac{v(t)}{i(t)} = \frac{80e^{-1000t}}{5 \times 10^{-3} e^{-1000t}} = 16 \text{ k}\Omega.$$

Then

$$L = \tau R = (1 \times 10^{-3})(1.6 \times 10^4) = 16 \text{ H}.$$

b) The energy dissipated in R from $t = 0$ to $t = t_1$ is

$$W = \int_0^{t_1} i^2(t) R dt, \quad t_1 = 0.5 \text{ ms}.$$

Thus,

$$W = R \int_0^{t_1} (5 \times 10^{-3})^2 e^{-2000t} dt = R (25 \times 10^{-6}) \int_0^{t_1} e^{-2000t} dt.$$

Evaluate the integral:

$$\int_0^{t_1} e^{-2000t} dt = \left[-\frac{1}{2000} e^{-2000t} \right]_0^{t_1} = \frac{1}{2000} (1 - e^{-2000t_1}).$$

Therefore,

$$W = R (25 \times 10^{-6}) \frac{1}{2000} (1 - e^{-2000t_1}).$$

Substitute $R = 1.6 \times 10^4$ and $2000t_1 = 2000(5 \times 10^{-4}) = 1$:

$$W = (1.6 \times 10^4)(25 \times 10^{-6}) \frac{1}{2000} (1 - e^{-1}) = 1.26424 \times 10^{-4} \text{ J} \approx 0.126 \text{ mJ}.$$